

Xiaozhen Ma

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EDUCATION

Southwest Jiaotong University and University of Leeds

Sep 2023 - Jul 2027

Bachelor of Engineering in **Electronic and Electrical Engineering** (Joint Program, dual degree from both universities)

University of California, Irvine

Jun 2026 - Aug 2026

UCInspire Summer Research Programme

St Catharine's College, Cambridge

Feb 2025

Interdisciplinary Winter School (Computer Science, Data Science, and Artificial Intelligence Track)

RESEARCH INTERESTS

Computer Vision, 3D Reconstruction, Robotic Perception, Autonomous Driving

RESEARCH EXPERIENCES

PIKAN-Drive: Progressive Physics-Informed KANs from PilotNet Steering Regression to BEV Trajectory Planning

Project Leader, Supervised by Prof. Fadi Kurdahi, UC Irvine

Jun 2026 - Present

- Developed a progressive PIKAN-Drive framework for end-to-end steering-angle regression based on PilotNet, incorporating MLP, KAN, and physics-informed KAN residual modules in a “visual feature extraction–baseline steering prediction–physical-state correction” architecture.
- Preprocessed and chronologically partitioned 45,406 sequential driving samples from the Udacity self-driving dataset, introduced buffer frames, and established a unified workflow for PilotNet training, testing, and model preservation.
- Trained and compared PilotNet, MLP, KAN and PI-KAN models under different physics-constraint weights, with steering smoothness, steering-rate and vehicle-state consistency constraints, and evaluated performance using MAE and RMSE.

LLM-Based Intelligent CAD Generation and Point-Cloud Surface Segmentation Optimisation

Research Assistant, Supervised by Prof. Dong-Ming Yan and Prof. Mingyang Zhao, National Laboratory of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences

Feb 2026 - Present

- Deployed and evaluated the tool-augmented CAD-Assistant vision-language model; analysed its workflow for task planning, code generation, geometric modelling and execution-feedback correction through the FreeCAD Python API; and conducted automated generation experiments across multiple CAD model categories.
- Optimised prompts for CAD sketch constraints and parametric modelling by incorporating dimensional relationships and geometric constraints, and analysed their effects on code-execution success, geometric validity and modelling accuracy.
- Reproduced the Point2CAD 3D point-cloud reconstruction pipeline and ParSeNet surface-segmentation model, including dataset preparation, pretrained-model testing, prediction generation and evaluation, and implemented data-interface compatibility between ParSeNet and Point2CAD.
- Analysed ParSeNet’s surface-instance segmentation mechanism based on dynamic graph convolution, point-feature embedding and mean-shift clustering. Proposed a three-axis projection boundary-detection and geometric-consistency method using multi-view projection, deployed the system on a dedicated server, trained ParSeNet from scratch and conducted comparative experiments on boundary constraints and loss weights.

Sparse Point Cloud 3D Railway Scene Reconstruction Technology Driven by Monocular Depth Estimation

Project Leader, Supervised by Prof. Zhao Tang, State Key Laboratory of Rail Transit Vehicle System

Apr 2025 - May 2026

- Developed the FrustumDense-SAM-XC3D railway-scene 3D reconstruction system by integrating multi-frame LiDAR point clouds, RGB images, and monocular depth information into a pipeline covering point-cloud densification, depth fusion, object segmentation, occlusion handling, and scene visualization.
- Proposed a multi-frame frustum back-projection densification method that projected LiDAR points from the current, historical, and future frames into the camera frustum for spatial aggregation, filling sparse regions around buildings, tracks, and trackside facilities to improve point-cloud coverage and structural completeness in large-scale scenes.
- Integrated DepthPro-based monocular metric depth estimation with LiDAR pose constraints and applied patch-wise fitting, boundary resampling, and moving least-squares prediction to correct scale errors in distant regions of the monocular depth maps and generate pixel-level dense 3D point clouds.
- Applied the pretrained SAM3 model to multi-frame image segmentation and object-boundary detection, used a tree-based data structure to manage nested objects and occlusion relationships, and implemented cross-frame co-visibility consistency checks for point-cloud matching, clustering, and unique object identification.
- Designed class-aware radius filtering and patch-based rendering algorithms to suppress accumulated noise from multi-frame point-cloud fusion, extract tracks, buildings, and transportation facilities by category, and visualize large-scale railway scenes for single-frame object extraction and railway asset presentation.

Machine Learning-Driven 3D Reconstruction of High-Speed Train Head Shapes

Project Leader, Supervised by Prof. Zhao Tang, State Key Laboratory of Rail Transit Vehicle System

Apr 2024 - Mar 2025

- Constructed a portable point cloud acquisition device incorporating multi-sensor fusion of LiDAR, cameras, and IMUs. Developed a radar-vision tightly coupled SLAM framework based on the open-source FAST-LIVO2 algorithm, with extended Kalman filtering employed to optimise camera pose calculations.

- Employed an enhanced ICP algorithm to refine registration accuracy, with a dynamic threshold denoising technique. Generated uniform, high-precision point clouds through normal vector estimation and point cloud resampling.
- Introduced an incremental k-d tree (IK-D-Tree) to enhance map storage and retrieval efficiency, and a dynamic object exclusion algorithm (based on motion consistency detection) to improve localisation robustness.
- Proposed a Tri-axial Projection Segmentation Interpolation and Reverse Hole Boundary Identification Method to achieve point cloud densification and uniform distribution in sparse point cloud scenarios caused by regional hole boundaries, elevating model integrity from 75% to 98%.
- Explored adaptive mechanisms for multiple algorithms in complex surface modelling, including Greedy triangulation, Poisson reconstruction, B-spline interpolation, Moving cuboid algorithm, and Radial basis function (RBF) interpolation.
- Independently authored the research report titled “Three-Dimensional Surface Reconstruction of High-Speed Train Noses from Sparse Point Clouds” based on the project, systematically summarizing the technical approach and comparing experimental results.

PROJECT EXPERIENCES

FPGA-Based Digital System Design

May 2026 - Jun 2026

- Developed functional modules for a hierarchical digital system using Quartus and a hardware description language. Connected modules through predefined ports, performed logic simulation, waveform analysis, and FPGA hardware testing.

STM32-Based Embedded Game Development

Apr 2026 - Jun 2026

- Collaborated in a three-person team to develop an embedded game using STM32 hardware and peripherals, contributing core game mechanics and an independent feature or level. Managed code with GitHub and documented development logs, system tests and demonstrations.

Audio-Visualisation PCB Design and Fabrication

Mar 2026 - May 2026

- Created the schematic and PCB layout for an audio-visualisation circuit in KiCad, using LEDs to indicate low-, mid- and high-frequency signals. Soldered electronic components, and used a multimeter for measurement and troubleshooting.

Emerging Optoelectronic System Design

Oct 2025 - Mar 2026

- Worked in an international team with University of Leeds students to research and design an optoelectronic system addressing a space-related challenge. Analysed its technical principles, commercial applications, environmental impact and engineering ethics, and produced an English-language design presentation video.

MCU Embedded Systems: Line-Following Model Car & Washing Machine Controller

Apr 2025 - May 2025

- Performed hardware programming of the line-following model car using Keil5, implementing functions such as LED display based on the STM32 microcontroller. Constructed a control circuit using MOSFETs as switches to control the model car's movement and turning. Designed and handcrafted the model car's body.
- Programmed the washing machine controller using Mbed to implement functions such as mode selection, child lock, time selection, pressure response, and light sensitivity based on the STM32 microcontroller.

HONORS AND AWARDS

- Second Prize (AI Innovation Track), 27th China Robot & Artificial Intelligence Competition - National Finals, 2025. (An Online Sports Course Evaluation System Based on Multi-Expert Collaborative Systems and Sentiment Analysis)
- Outstanding Student Leader, SWJTU-Leeds Joint School, Southwest Jiaotong University, 2025.
- Mingcheng Award, SWJTU-Leeds Joint School, Southwest Jiaotong University, 2024.

EXTRACURRICULAR ACTIVITIES AND VOLUNTEER

Academic Innovation Base of SWJTU-Leeds Joint School, Southwest Jiaotong University

Sep 2023 - Present

Roles: Member (2023-2024); Head of Activities (2024-2025); President (2025-Present)

- As president, coordinated over 30 members and 4 departments, liaised with more than 80 guests, planned event schedules, venues, and themes, and spearheaded the development of the “Leeds Tech Month” flagship competition, attracting over 500 participating teams, with events reaching more than 3,000 faculty and students.

TECHNICAL SKILLS AND ADDITIONAL INFORMATION

Technical Skills: C/C++/Python/MATLAB(VS2017, MbedStudio, OnlineGDB, Keil5); ROS2; Linux; SLAM (FAST-LIO/FAST-LIVO2/R3LIVE); point-cloud/3D reconstruction; VTK/PCL/OpenCV; embedded integration (STM32).

Languages: Mandarin(Native), English(Fluent). **Interests:** Eight Brocades, Simplified 24-Form Tai Chi Chuan